Research Outlook: The Geometry of Continual Self-Learning

Allen Schmaltz
Reexpress AI
allen@re.express

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Abstract

Robust, interpretable estimators of the predictive uncertainty over neural networks are hypothesized to be key for constructing continual *self*-learning systems, and ensuring that human operators remain meaningfully in control of those systems. Concretely, SIMILARITY-DISTANCE-MAGNITUDE (SDM) estimators make use of the geometry of the feature-representation space of neural models to robustly estimate the high-probability region of the output distribution, unlocking the potential for automated self-labeling. The explicit constraints against the support set prospectively provide a guard against degenerate mislabeling over training cycles, and the approximately conditional partitioning of the distribution and instance-wise mapping to the support set enable efficient active learning. We briefly introduce this research agenda, which will require advances in multiple testing, training dynamics, and catastrophic forgetting.

1 Introduction

Schmaltz (2026b) introduces an approach for uncertainty quantification over neural networks, with a particular emphasis on small $|\mathcal{Y}|$ classification, as needed for binary classification of instruction-following (Schmaltz, 2026a). SIMILARITY-DISTANCE-MAGNITUDE (SDM) estimators use the geometry of the feature-representation space of the model against the training (or support) set to estimate a region of the distribution for which the class- and prediction-conditional accuracy is at least α, a value near 1 (e.g., $\alpha = 0.95$). This is referred to as the HIGH-RELIABILITY (HR) region and is intended for selective classification settings for which reliable, robust, and interpretable estimates in the high-probability region are paramount.[1]

SDM estimators seek informative (sharp) estimates over in-distribution data, while preferring conservative estimates rather than mis-calibration over covariate-shifted and out-of-distribution data. For datapoints falling outside the HR region, we can rank the relative probability of the remaining instances by running the partitioning algorithm for descending α values, successively excluding the datapoints of each higher region that is found. With such *nested* estimates over the residuals, instances falling in regions partitioned by higher α values are preferable, other things being equal. This then also provides a natural notion of out-of-distribution, as any point not assigned to a nested region is effectively out-of-distribution to the estimator.

SDM estimators are a substantive departure from existing approaches for estimating uncertainty over language models for at least three key reasons. First, the targeted quantity of interest, class- and prediction-conditional accuracy at least α, is a more stringent criteria than typical marginal estimators, but required for real-world tasks such as binary classification of instruction-following where the test-time distribution can be imbalanced and/or of differing difficulty across classes. Second, SDM estimators are unusually robust with respect to covariate shifts, as growing evidence suggests that HR regions with α values near 1 exhibit improved robustness to covariate shifts in natural language data relative to alternatives that marginalize over these metric-learning signals of the epistemic (reducible) uncertainty, as high SIMILARITY (q) values (an estimate of depth-wise neighborhood prediction and label matches into the support set) and DISTANCE quantiles (d, which is the minimum, across classes, 1-quantile of the L^2 distance to the nearest support set match, where the minimum is a

[1] SDM estimators conclude my publicly stated research agenda from December 2020: "*From* Parameter Identification *to* Introspection, Constrained Inference, & Updatability: Toward An Actionable Understanding of Neural Networks" (https://raw.githubusercontent.com/allenschmaltz/research/main/research_statement/research.pdf).

guard against class-wise skew in the distribution of outliers) near 1 serve as relatively stable partitions of the distribution. Finally, SDM estimators mark a substantive departure in how we approach interpretability of neural networks, shifting the locus of interpretability *from* trying to make sense of statistically non-identifiable parameters *to* understanding the estimators of the predictive uncertainty. They bring principled uncertainty quantification to the established notion of interpretability-by-exemplar (Schmaltz, 2021), providing a mapping between a test instance and the training, or support, set to trace the provenance of a prediction and the associated signals that determined the estimate of the predictive uncertainty.

2 Toward Continual *Self*-Learning

Absent a robust, approximately conditional estimator of the predictive uncertainty, progress in true continual self-learning (a.k.a., recursive self-improvement, and related) is unlikely, as the learning process will become degenerate over time as covariate shifts and out-of-distribution inputs derail the self-annotation of new data. Even if such annotation is limited to "verifiable" (i.e., deterministic, symbolic, or formally provable) cases, the learning process still needs to route to such verification, and such routing is a probabilistic task.

SDM estimators thus provide a key missing piece beyond simply scaling existing models and running them in a naive loop with external tools. Critically, the interpretability-by-exemplar properties of SDM estimators ensure that human operators remain in control of the self-learning process, providing meaningful post-hoc auditing and the ability to make local updates by strategically changing the support set.

SDM estimators are a parsimonious algorithm for calculating the uncertainty of neural networks. The goal is to find an analogously parsimonious algorithm for the next level of abstraction, which is the aforementioned self-learning process. At a high-level, the thinking is that we need to minimize the SDM loss[2] (c.f., cross-entropy loss) to create (and ensure) a self-consistent system, with respect to the available data, over time.

This will prospectively involve new work analyzing and addressing the following areas:

- **Multiple-testing.** Is simply composing SDM estimators over time sufficient, or do we need a correction (e.g., Bonferroni corrections and related)?
- **Learning dynamics.** How do the geometric constraints, namely the SIMILARITY and DISTANCE, evolve across multiple rounds of self-annotated learning?
- **Catastrophic forgetting.** With SDM estimators, we can, in principle, address catastrophic forgetting by simply conservatively branching across copies of a model that have been locally updated. However, prospectively, the compute and storage costs become prohibitive when naively constructing such a search graph over time. How can we effectively distill such a search graph into a single model without inducing catastrophic abstention (in the selective classification sense) of previously learned information, and/or recently learned information, from the test-time estimator? And more broadly, what is the precise relationship between forgetting and abstention with an estimator that has explicit controls on the epistemic uncertainty?

[2] See Eq. 7 in Schmaltz (2026b).

References

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